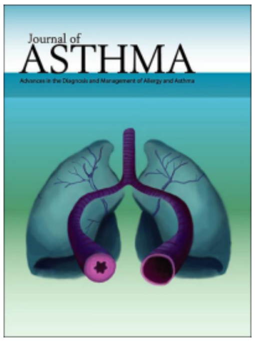

Journal of Asthma

ISSN: 0277-0903 (Print) 1532-4303 (Online) Journal homepage: www.tandfonline.com/journals/ijas20

An Evaluation of Standardizing Target Ventilation for Eucapnic Voluntary Hyperventilation Using FEV_1

Barry A. Spiering, Daniel A. Judelson & Kenneth W. Rundell

To cite this article: Barry A. Spiering, Daniel A. Judelson & Kenneth W. Rundell (2004) An Evaluation of Standardizing Target Ventilation for Eucapnic Voluntary Hyperventilation Using FEV_1, Journal of Asthma, 41:7, 745-749, DOI: 10.1081/JAS-200028004

To link to this article: https://doi.org/10.1081/JAS-200028004

JOURNAL OF ASTHMA
Vol. 41, No. 7, pp. 745–749, 2004

ORIGINAL ARTICLE

An Evaluation of Standardizing Target Ventilation for Eucapnic Voluntary Hyperventilation Using FEV_1

Barry A. Spiering, M.S.,[1] Daniel A. Judelson, M.A.,[1] and Kenneth W. Rundell, Ph.D., F.A.C.S.M.[2,*]

[1]University of Connecticut, Storrs, Connecticut, USA
[2]Marywood University, Scranton, Pennsylvania, USA

ABSTRACT

Athletes are required to provide objective documentation of exercise-induced bronchoconstriction (EIB) to use β_2-agonists during Olympic competition. A positive response to bronchial provocation by eucapnic voluntary hyperventilation (EVH) is considered acceptable confirmation of EIB. Thirty times forced expiratory volume in the first second (FEV_1) is recommended as EVH target ventilation (TV), an intensity intended to estimate 85% of maximal voluntary ventilation (MVV). There is a paucity of data examining the accuracy of predicting MVV from FEV_1 in elite athletes. The purpose of this study was to evaluate the efficacy of $30 \times FEV_1$ as standardized EVH TV. Maximal minute ventilation during exercise (VEmax) and pulmonary function of 78 elite winter athletes (25 males, 53 females; 25 EIB positive, 53 normal) were analyzed retrospectively. Adequacy and variability of the equation was ascertained by examining the ratio of EVH TV ($30 \times FEV_1$) to VEmax. VEmax was $99 \pm 11\%$ of predicted MVV ($35 \times FEV_1$) and was positively related ($r = 0.85$, $p \leq 0.05$). TV was $88 \pm 9\%$ of VEmax (range: 64–109). For elite athletes, the high variability in $30 \times FEV_1$ to standardize TV for EVH may result in under-diagnosis for low-end outliers. Since VEmax of elite endurance athletes is typically known (via maximal aerobic testing) we recommend 85% VEmax as a more accurate and reliable method to establish EVH TV for this group; if VEmax is not available, then 85% of measured MVV may be used.

Key Words: Asthma; Bronchospasm; Exercise-induced bronchoconstriction; Isocapnic hyperpnea; Olympics.

*Correspondence: Kenneth W. Rundell, Ph.D., F.A.C.S.M., Professor of Health Science, Director of the Human Performance Laboratory, Keith J. O'Neill Center for Healthy Families, Marywood University, 2300 Adams Ave., Scranton, PA 18509, USA; E-mail: rundell@marywood.edu.

DOI: 10.1081/JAS-200028004
0277-0903 (Print); 1532-4303 (Online)
www.dekker.com

INTRODUCTION

In November 2001, the Medical Commission of the International Olympic Committee (IOC-MC) updated its policy on the use of β_2-agonists as prophylaxis for EIB during Olympic competition (1). The new policy required spirometric tracings to document the presence of EIB. This update was implemented in response to the reported high prevalence of EIB in elite cold-weather athletes (2–7) and a trend of increased athlete use of β_2-agonists [11.2% in the 1984 Summer Games (8); 15.3% in the 1996 Summer Games (9)].

The updated IOC-MC policy designated EVH as the preferred challenge to diagnose EIB. EVH was chosen because it is a potent bronchoconstrictor, symptomatically similar to exercise, and relatively simple to standardize. The EVH protocol involves: 1) determination of the subject's baseline forced expiratory volume in the first second (FEV_1); 2) hyperventilation of a 5% carbon dioxide dry gas mixture for six minutes at a rate equal to 30 times the subject's FEV_1 ($30 \times FEV_1$); and 3) measurement of post-challenge FEV_1 at 0, 5, 10, and 15 minutes post-test. A $\geq 10\%$ fall in FEV_1 from pre- to postchallenge is considered positive for EIB (EIB+) (10).

The EVH TV ($30 \times FEV_1$) is intended to simulate 85% of predicted maximal voluntary ventilation (MVV). As athletes typically employ 60–90% of MVV during exercise, 85% of MVV is considered a demanding but sustainable intensity (10). Although predictions of MVV based on $35 \times FEV_1$ have been reported accurate and reliable for groups (11), several reports (11–14) indicate that FEV_1-based predictions of MVV show high individual variability. Harber et al. (11) concluded, ''the ratio of MVV to FEV_1 was too variable to be of use in determining whether an individual's results are reliable'' (i.e., predictions of MVV based on FEV_1 lead to significant individual error). This variability is important because fluctuations in TV can alter the bronchoconstrictive response (15).

Furthermore, subjects in previous studies examining FEV_1-based prediction equations used sedentary normals, asthmatics, or patients with COPD. FEV_1-based predictions of MVV are untested in the elite athlete, further questioning the validity of their use in a standardized test for evaluating Olympic athletes. Therefore, the purpose of this study was to evaluate the accuracy and inter-subject variability of using $30 \times FEV_1$ as a standardized TV for EVH testing of elite athletes.

METHODS

Subjects

Subjects were 78 elite winter athletes (25 males, 53 females). All had previously performed maximal aerobic capacity (VO_2max) tests and had been evaluated for EIB via field-based exercise at the United States Olympic Training Center Sport Physiology Laboratory in Lake Placid, NY. Of the 78 subjects, 32% (12 males, 13 females) were EIB+ and 68% (13 males, 40 females) had normal airway response to exercise. Prior to data collection, all subjects signed a written informed consent and experimental procedures were approved by the Institutional Review Board of the Coaching and Sport Science Division of the United States Olympic Committee.

Procedures

FEV_1 and maximal ventilation during exercise (VEmax; one minute rolling average) data were obtained by retrospective analysis of resting pulmonary function and sport-specific VO_2max tests (running, cycling, roller-skiing, or roller-blading). Resting FEV_1 was selected from the best-of-three baseline maneuvers as defined by the highest sum of FVC and FEV_1. Criteria used to select VEmax included a plateau in VO_2 with increasing workloads and/or volitional fatigue, which ensured achievement of VO_2max and VEmax.

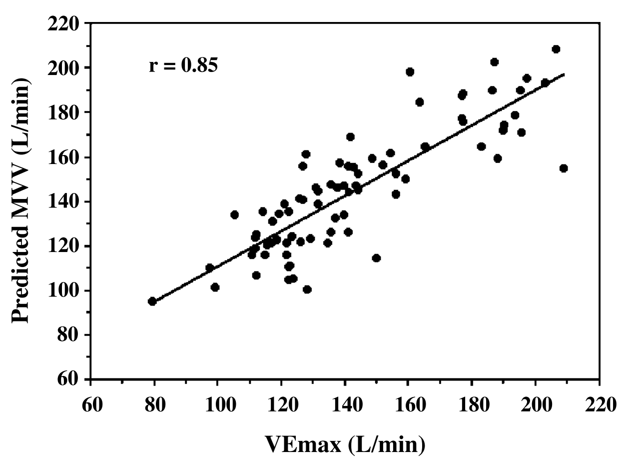

Figure 1. Relationship between VEmax and predicted MVV. Trend line (solid) is shown. Measurement variables were significantly correlated (r = 0.85; $p \leq 0.05$).

Table 1. EVH TV/VEmax ratio for groups based on gender and EIB status.

		Males	Females	Total
EIB −	n	12	41	53
	Mean ± SD	86 ± 9	89 ± 10	89 ± 9
	Range	73–106	65–109	65–109
EIB +	n	13	12	25
	Mean ± SD	84 ± 9	87 ± 10	86 ± 9
	Range	64–96	73–105	64–109
Total	n	25	53	78
	Mean ± SD	85 ± 9	89 ± 10	88 ± 9
	Range	64–106	65–109	64–109

EIB: exercise-induced bronchoconstriction; EVH: eucapnic voluntary hyperventilation; VEmax: maximal exercising ventilation; TV: target ventilation.

Statistical Analyses

Predicted MVV ($35 \times FEV_1$) was compared to VEmax using linear regression. EVH TV ($30 \times FEV_1$) was normalized to individual VEmax, which provided an index of relative intensity for EVH TV. Descriptive statistics (mean, median, standard deviation, interquartile range, 10th to 90th percentile range, and 95% confidence intervals [CI]) were calculated for predicted MVV, VEmax regression and normalized EVH TV. Data are presented as mean ± SD. Independent t-tests were used to examine differences in normalized EVH TV between gender and/or EIB status. Significance was set at $p \leq 0.05$.

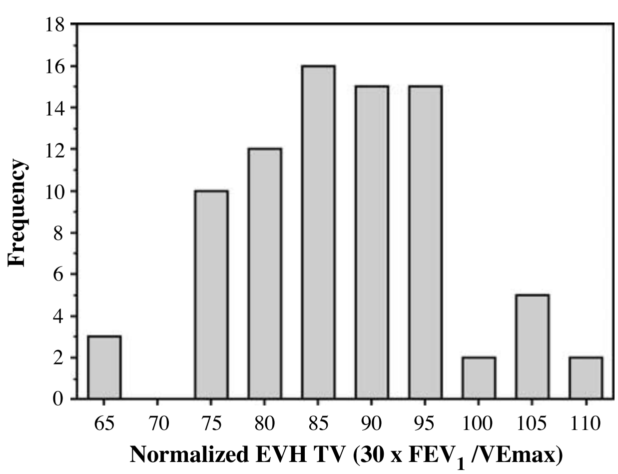

Figure 2. Histogram of normalized EVH TV ($30 \times FEV_1$/VEmax). Data were normally distributed (skewness = −0.10, ns; kurtosis = 0.09, ns) and demonstrated a number of individuals below and above the optimal ventilation rate for EVH.

RESULTS

VEmax was 99 ± 11% of predicted MVV (Range: 79–135% predicted MVV). There was a significant positive correlation between VEmax and predicted MVV (Fig. 1; r=0.85; $p \leq 0.05$). Three subjects (4%) fell outside of the 95% confidence interval (mean ± 2 SD; CI: 77–121) and all were above the CI.

Table 1 depicts EVH TV relative intensity ($30 \times FEV_1$/VEmax) for groups based on gender and EIB status. No differences existed between groups;

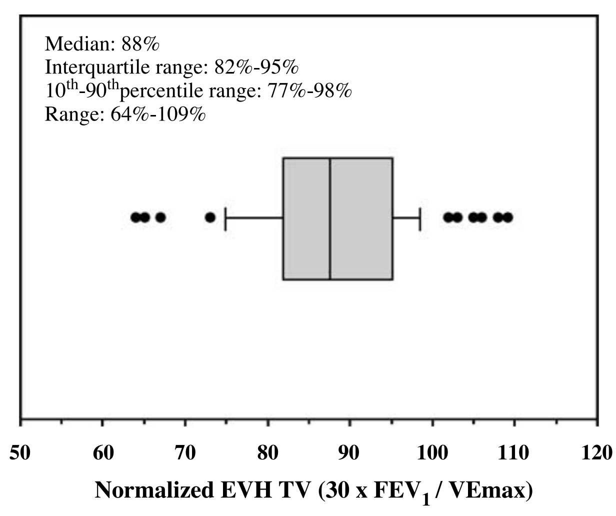

Figure 3. Dispersion of EVH TV ($30 \times FEV_1$) after normalizing data to individual VEmax. The center vertical line represents the median; the boxed region indicates the interquartile (I.Q.) range; lines extending from the box indicate the 10th to 90th percentile (10th–90th%) range; dots indicate values lying outside the 10th–90th percentile range.

therefore, subjects were pooled for analysis. Data were normally distributed (Fig. 2).

Relative intensity of the EVH TV was 88±9% VEmax (Range: 64–109% VEmax). Six subjects (8%) fell outside the 95% CI (70–106). Of these six subjects, values were equally split above and below the CI. Figure 3 displays the dispersion of values according to the median (88), interquartile (IQ) range (82–95), and 10th–90th percentile range (77–98). Fourteen (18%) subjects fell outside the 10th to 90th percentile range; of these, values were equally split above and below the 90th and 10th percentiles.

DISCUSSION

This study is consistent with work done by others on nonathlete populations that demonstrated ventilation predictions based on FEV_1 are accurate for subject populations, but not individuals (11–14). Our results indicate that the EVH TV prediction equation ($30 \times FEV_1$) provides a mean intensity of 88±9% of VEmax with normally distributed values. However, because the results of EVH are dependent upon TV (15), the individual variability in our data (range=64–109% of measured VEmax) suggests that the use of TV predictions based on FEV_1 may limit the diagnostic precision of EVH.

Little data on ventilatory reserve of elite athletes is available. Anderson et al. (10) suggested that normal individuals rarely exceed 60% of predicted MVV and that elite athletes rarely exceed 90% of predicted MVV during exercise. In normal subjects (20 to 80 years), Blackie et al. (16) found that VEmax was 61±14% of predicted MVV. Folinsbee et al. (17) reported that sedentary males ventilated at 71% of measured MVV during maximal exercise, while elite males cyclists used 89% of measured MVV during exercise. VEmax measured in our subjects was 99±11% of predicted MVV. The elite athletes in our study were able to minimize their ventilatory reserve during exercise and may require a high TV for an appropriate response to EVH.

The EVH TV equation produced 6 values outside the 95% CI. ''Low-end'' outliers tested by EVH at a TV of $30 \times FEV_1$ would ventilate at an inappropriately reduced intensity. This may lead to a false negative diagnosis due to a TV insufficient to evoke an EIB response. O'Cain et al. (15) have shown that the bronchoconstrictive response to EVH is directly proportional to TV. Furthermore, Rundell et al. (18) suggested that elite athletes require a high exercise intensity to elicit a bronchoconstrictive response. Therefore, the variability in EVH TV may manifest itself in underdiagnoses of low-end outliers and, ultimately, prevent EIB+ Olympic athletes from receiving necessary and appropriate medication during competition.

According to Anderson et al. (10), EVH TV standardization using $30 \times FEV_1$ ''is recommended as first approach.'' However, the authors state, ''... the test could theoretically be carried out in a manner that best simulates their sporting event i.e. the duration, ventilation, and temperature can be changed according to the conditions of their sport.'' Although EVH procedures allow for flexibility (i.e., duration, intensity, temperature), the protocol should be accurately and reliably standardized for the following reasons: 1) it is difficult to simulate actual exercise duration and intensity in some Olympic sports (e.g., the repeated interval nature of ice hockey, the long duration of cross-country skiing); 2) many team physicians, allergists, and/or pulmonologists may not have the equipment, means, or time to complete a field-based exercise challenge or the knowledge or ability to accurately simulate the demands of an athlete's activity; and 3) EVH retesting for verification of treatment efficacy requires a repeatable TV. Therefore, establishing a standardized EVH TV of sufficient intensity will enhance the accuracy and reliability of the EVH challenge.

As demonstrated, FEV_1-based predictions for EVH TV were inappropriately low for approximately 40% of our subjects, and we contend that a more rigorous approach to determining EVH TV is necessary. Since many Olympians and Olympic-hopefuls are regularly tested for VO_2max, individual VE_{max} data is typically available. Therefore, we recommend using 85% of VEmax as a standardized VR for EVH. This intensity should be adequate for EIB provocation, yet maintainable for the duration of the EVH challenge. If VEmax data is not available, EVH TV should be 85% of measured MVV.

In conclusion, the success of EVH as an EIB challenge test will be dictated by its standardization. Although the EVH TV prediction equation ($30 \times FEV_1$) is accurate for a large study group population, it instilled a wide range in relative TV intensity, reducing diagnostic potential. Future work should compare the sensitivity of EVH at $30 \times FEV_1$ to EVH at 85% VEmax and/or 85% measured MVV.

REFERENCES

1. Schamasch P. $Beta_2$ adrenoceptor agonists and the Olympic Winter Games in Salt Lake City, 2001.

Available at http://multimedia.olympic.org/pdf/en_report_20.pdf (accessed Nov 2001).
2. Mannix ET, Farber MO, Palange P, Galassetti P, Manfredi F. Exercise-induced asthma in figure skaters. Chest 1996; 109(2):312–315.
3. Provost-Craig MA, Arbour KS, Sestili DC, Chabalko JJ, Ekinci E. The incidence of exercise-induced bronchospasm in competitive figure skaters. J Asthma 1996; 33(1):67–71.
4. Rundell KW, Jenkinson DM. Exercise-induced bronchospasm in the elite athlete. Sports Med 2002; 32(9):583–600.
5. Rundell KW, Im J, Mayers LB, Wilber RL, Szmedra L, Schmitz HR. Self-reported symptoms and exercise-induced asthma in elite athletes. Med Sci Sports Exerc 2001; 33(2):208–213.
6. Sue-Chu M, Larsson L, Bjermer L. Prevalence of asthma in young cross-country skiers in central Scandinavia: differences between Norway and Sweden. Respir Med 1996; 90(2):99–105.
7. Wilber RL, Rundell KW, Szmedra L, Jenkinson DM, Im J, Drake SD. Incidence of exercise-induced bronchospasm in Olympic winter sport athletes. Med Sci Sports Exerc 2000; 32(4):732–737.
8. Voy RO. The U.S. Olympic Committee experience with exercise-induced bronchospasm, 1984. Med Sci Sports Exerc 1986; 18(3):328–330.
9. Weiler JM, Layton T, Hunt M. Asthma in United States Olympic athletes who participated in the 1996 Summer Games. J Allergy Clin Immunol 1998; 102(5):722–726.
10. Anderson SD, Argyros GJ, Magnussen H, Holzer K. Provocation by eucapnic voluntary hyperpnoea to identify exercise induced bronchoconstriction. Br J Sports Med 2001; 35(5):344–347.
11. Harber P, SooHoo K, Tashkin DP. Is the MVV:FEV1 ratio useful for assessing spirometry validity? Chest 1985; 88(1):52–57.
12. Mohan-Kumar T, Gimenez M. Maximal ventilation at rest and exercise in patients with chronic pulmonary disease. Respiration 1984; 46(3):291–302.
13. Dillard TA, Piantadosi S, Rajagopal KR. Prediction of ventilation at maximal exercise in chronic air-flow obstruction. Am Rev Respir Dis 1985; 132(2):230–235.
14. Fairshter RD, Carilli A, Soriano A, Lin J, Pai U. The MVV/FEV1 ratio in normal and asthmatic subjects. Chest 1989; 95(1):76–81.
15. O'Cain CF, Dowling NB, Slutsky AS, Hensley MJ, Strohl KP, McFadden ERJ, Ingram RHJ. Airway effects of respiratory heat loss in normal subjects. J Appl Physiol 1980; 49(5):875–880.
16. Blackie SP, Fairbarn MS, McElvaney NG, Wilcox PG, Morrison NJ, Pardy RL. Normal values and ranges for venitlation and breathing pattern at maximal exercise. Chest 1991; 100(1):136–142.
17. Folinsbee LJ, Wallace ES, Bedi JF, Horvath SM. Exercise respiratory pattern in elite cyclists and sedentary subjects. Med Sci Sports Exerc 1983; 15(6):503–509.
18. Rundell KW, Wilber RL, Szmedra L, Jenkinson DM, Mayers LB, Im J. Exercise-induced asthma screening of elite athletes: field versus laboratory exercise challenge. Med Sci Sports Exerc 2000; 32(2):309–316.